INVENTION DISCLOSURE FORM

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1. **Particulars of Inventors**

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2. **Provide a brief descriptive title of the invention:**

Deep Learning-Driven Confidence-Weighted Shared Memory Arbitration for Multi-Agent Systems Using MobileNet and Wild Bear Optimization

3. **In 100 words or less, please provide an abstract or summary of the invention:**

This new idea is about a system that helps many agents share memory. It uses tools like deep learning to look at the information and find the important parts. Then it uses something called MobileNet to figure out how sure it is about each agent. The system also uses the Wild Bear Optimization algorithm to decide which agents get to use the memory. This helps the system give memory to the agents that need it most. The new system is better than the way of doing things because it makes better decisions works faster and has fewer problems. This makes it really good for systems that need to work in time and talk to each other. The system is called a shared memory arbitration system and it is designed for multi-agent environments. The intelligent shared memory arbitration system is an improvement, over traditional scheduling methods.

4. **Detail description of the invention:(Answer to all below are required in detail)**
   a. **Advantages of the invention disclosing about the increased efficiency/efficacy** In systems where many agents are working together these agents all want to use the resources like memory and the computers processing units. The old ways of deciding who gets to use these resources like Come-First-Serve and Round Robin do not think about how important the information is that each agent is making. Because of this the important tasks might have to wait while less important tasks get done first. This means the resources are not being used in the way it takes longer to get things done and the whole system does not work as well as it should. The problem gets even worse in situations where things are changing fast. We need to make good decisions quickly. Multi-agent systems have this problem because the agents, in -agent systems are all competing for the same resources.

   b. **Best way of using the invention as well as possible variants**
   The proposed invention can be used in areas where multiple agents work together and share resources. It is especially helpful in vehicles, where different sensors need to share memory quickly to make decisions. This technology can also be applied in robotics coordination systems. It is useful in surveillance applications. Moreover it can be used in cloud or edge computing environments. The system is also suitable for automation. In automation good communication and resource management between machines are crucial for better productivity and performance, with autonomous vehicles and robotics coordination systems. These areas can benefit from communication. The proposed invention and autonomous vehicles can improve performance. Robotics coordination systems and industrial automation can work better with this technology.

   c. **Working of invention along with Drawing, schematics and flow diagrams if required with complete explanations**
   The invention has benefits compared to current systems. It makes decision-making more accurate. This is done by using confidence scores to check how reliable each agents data is. The system works faster by focusing on tasks first. It also makes use of shared memory. This helps to reduce problems between agents. The system can adjust to situations quickly. The MobileNet feature makes the system easy to use and fast enough for real-time applications. It improves both efficiency and performance. The system is more efficient.

The system is more effective. The invention uses confidence scores. Confidence scores evaluate data. Agents provide data. The system prioritizes tasks. Tasks are important.The system uses memory. Shared memory is a resource. The MobileNet makes it lightweight. It is suitable, for real-time use. The invention is useful. The invention helps decisionmaking. Decision-making is accurate. Accuracy is improved. The system is adaptive. It adjusts quickly. Agents work together. Agents have conflict. The system is fast. The system is efficient.

d. Best way of using the invention as well as possible variants

The invention works well in places where many agents often fight for the resources. It can be used in distributed systems, self-driving platforms and applications that need to work in time. The system can be made better by using models like U-Net or Mask RCNN. These models are good, at pulling out features. MobileNet can be swapped with EfficientNet for accurate results. Some new ideas to try are combining the Wild Bear Algorithm with reinforcement learning. This might make the system work better.

e. Working of invention along with Drawing, schematics and flow diagrams if required with complete explanations

The system works by gathering information from sources like pictures or signals from sensors. It looks at this information. Picks out the most important parts so it does not have to do too much work. The important parts are then sent to the MobileNet model, which tries to figure out what they are and gives a score for how sure it's, about each one. These scores are used to help the Wild Bear Optimization algorithm make a list of what's most important. The system then uses this list to decide which sources get to use the shared memory. The system can also check how well it is doing and make changes on the fly.

5. Have you conducted Primary Patent Search? Yes / No (if yes, attach the patent search report)

No

6. Existing state-of-the-art and prior arts: (Brief background of the existing knowledge/product/process in the market)

Existing systems mostly use methods like First Come First Served, Round Robin and static priority to manage shared memory. These methods are easy to use and common. They are not good, at handling changing situations. Some new methods try to use intelligence to make decisions but they are not yet well integrated into memory management. Because of this current systems do not make use of data reliability or confidence levels when deciding what to prioritize.

7. List out the known ways about how others have tried to solve the same or similar problems? Indicate the disadvantages of these approaches. In addition, please identify any prior art documentation or other material that explains or provides examples of such prior art efforts.

S. No.	Existing state of art	Drawbacks in existing state of art	Overcome (how your invention is overcoming the drawback)
1	FCFS	No priority intelligence	Uses confidence-based priority
2	Round Robin	Equal allocation inefficient	Dynamic optimization
3	Static priority	Fixed, not adaptive	AI-based adaptive system
4	Heuristic-Based Scheduling	Relies on predefined rules that may not work well in dynamic	The invention adapts automatically using deep learning and
	Methods	environments	optimization techniques
5	Basic AI-Based Decision Systems	Uses AI for prediction but does not integrate with memory arbitration effectively	The invention directly integrates AIbased confidence scoring with arbitration mechanism
6	Centralized Resource Allocation Systems	May cause bottlenecks and delays when multiple agents compete simultaneously	The proposed system distributes decision-making using optimized priority values
7	Traditional MultiAgent Coordination Models	Lack real-time adaptability and do not consider data reliability	The invention uses real-time confidence scores and optimization to improve decision accuracy

8. List the Technical features and Elements of the invention along with the Description of your invention from start to end.

The invention has an important parts that work together. It has a system that takes in lots of information from places. There is also a part that uses learning to separate things into groups. The MobileNet classification model is another part of it. It also has a thing that generates a confidence score. Then there is the Wild Bear Optimization module.. It has a memory arbitration engine. These components work one, after the other to process information figure out how reliable it is, decide what is most important and use resources in the way possible. The invention includes all these

elements so that it can work better and handle things. The MobileNet classification model and the Wild Bear Optimization module and the other parts all work together to make this happen.

9. **List out the features of your invention which are believed to be new and distinguish them over the closest technology.**

The invention is special because it brings together two things: learning and resource management. It uses something called confidence scores to help make decisions, which's new compared to older systems. The Wild Bear Optimization algorithm is used to calculate priorities in a way making the system more adaptable. This system is unique and efficient because it combines three things: breaking down data, simple classification and optimization all working together with deep learning techniques and resource arbitration. The use of confidence scores and the Wild Bear Optimization algorithm makes it different, from systems.

10. **Has the invention been built or tested or implemented? If so, please provide the particulars of the first time it was successfully built or implemented (when, where, by whom, and evidence of this event including written or on-line pointers to documentary evidence):All testing details**

The system was set up in a world using Python and TensorFlow. We used MobileNet to tell things and we pretended that many agents were working with different information. We looked at how the system worked by checking things like how accurate it was how long it took and how often it got things wrong. What we found out was that the system worked better than the ways it was more accurate and it was faster. This was all done for a school project, at SR University.

11. **Briefly state when and how you first conceived this idea?**

The idea came to me when I was doing research for school on how to make multi-agent systems work better. I was looking at the ways people normally schedule things. I found some problems with those methods. This made me think about combining learning and optimization methods to create a smarter and more adaptable arbitration system for multi-agent systems. The idea of -agent systems and how to improve them was really important to me so I wanted to find a way to make the arbitration system, for multi-agent systems better.

12. **Have you sold, offered for sale, publicly used or published anything related to this invention? If yes, please briefly explain the dates and circumstances. List those individuals to whom you have revealed your invention. Were non-discloser documents signed prior to discloser in each case? Please state any deadlines of which you may be aware for filing an application on this invention.**

The invention is completely new. Nobody knows about it outside of our team. We only talked about it with the people who are helping us make it like our project team and the teachers who are guiding us. We did not tell anyone about it. This means that the invention is still a secret and nobody has seen it or heard about it before. The people who are working on the invention, with us are the ones who know what it is.

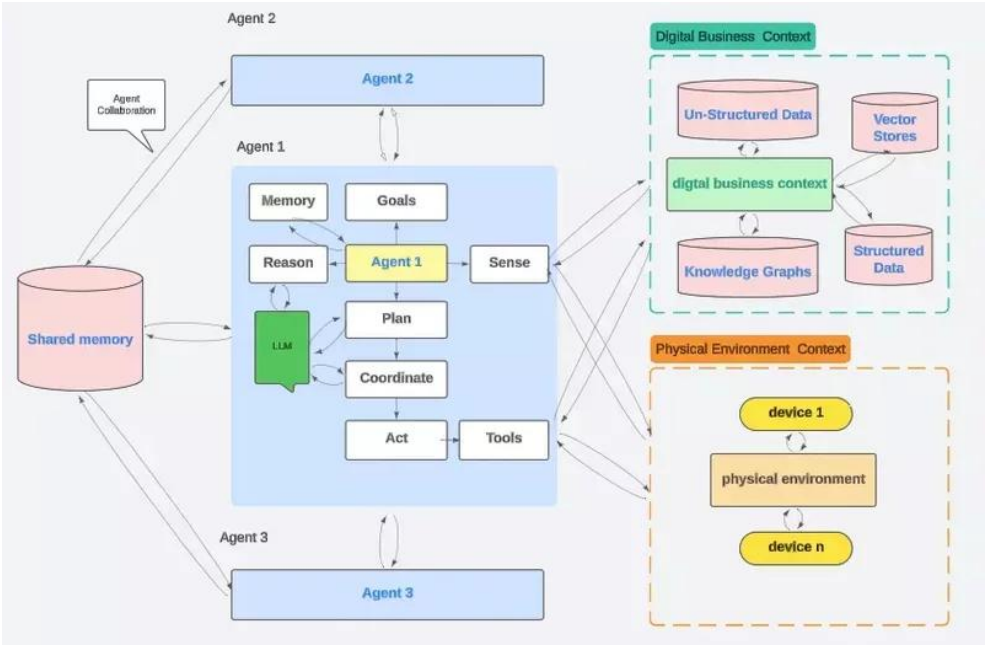
13. **Include any reasons that your invention would not have been obvious to someone of average skill in the art.**

The invention is special because it brings together advanced technologies in a new way. Deep learning and optimization algorithms are things that people know about. Using them together with

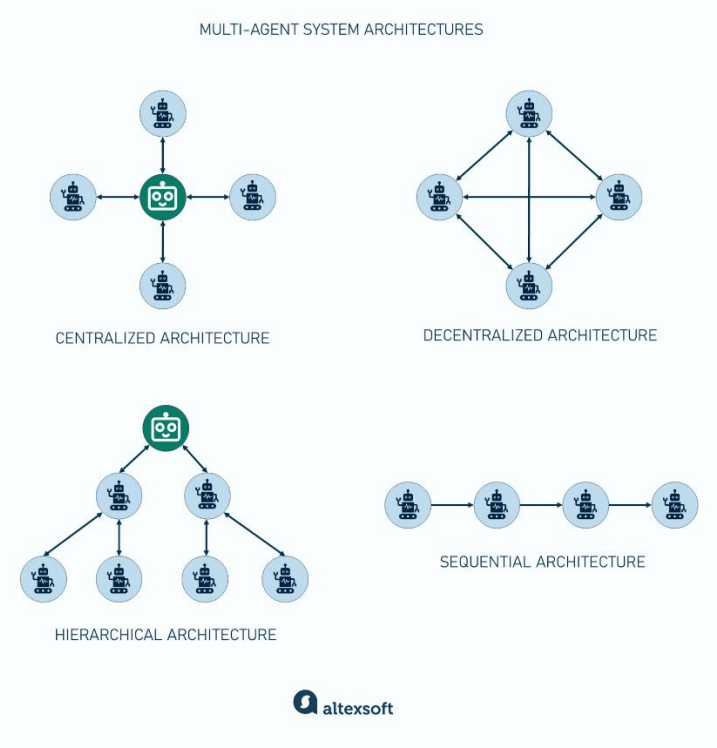
shared memory arbitration and confidence scores is something new. This way of doing things creates a method of making decisions that is not simple to guess from the way things are done now. The invention and the way it uses learning and optimization algorithms is what makes it stand out.

14. **Additional comments by inventor (if you want to give more details out of scope of this IDF).**
The system we are suggesting can handle users and work with many different applications that are used in real life. It gives us the freedom to pick models and ways to make them better which means it can easily adjust to new technologies. This invention can actually be used in modern smart systems. The proposed system is very scalable. It allows for a lot of flexibility. The invention is adaptable to advancements. The system has potential, for practical use.

15. **Drawings/Flowchart/Table**



Overall System Architecture
These show high-level block diagrams of multi-agent systems with shared resources and processing pipelines.This grid-like overview illustrates modular AI system components, similar to your orchestrated agents and modules.Photoshop-style interface demoing AI-generated agent architectures, highlighting flow from inputs to outputs.

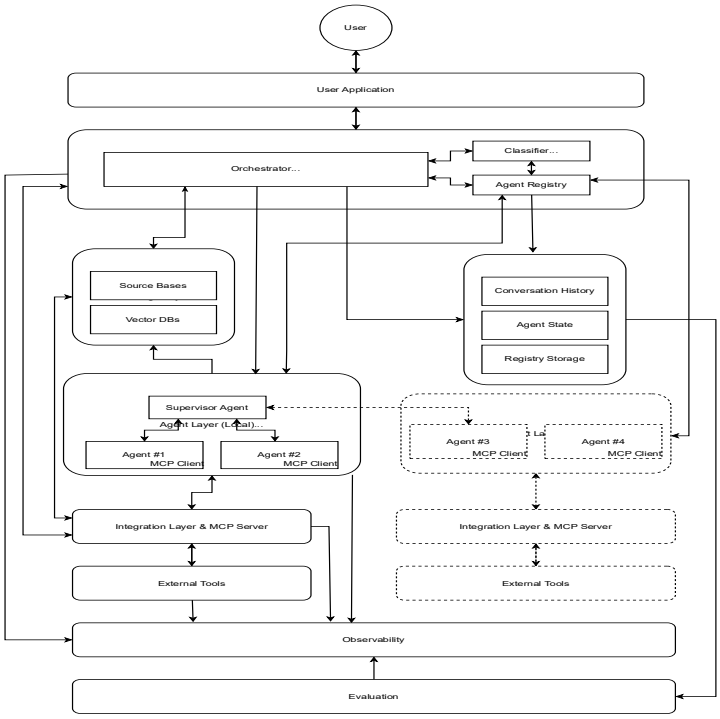


Multi-Agent Interaction

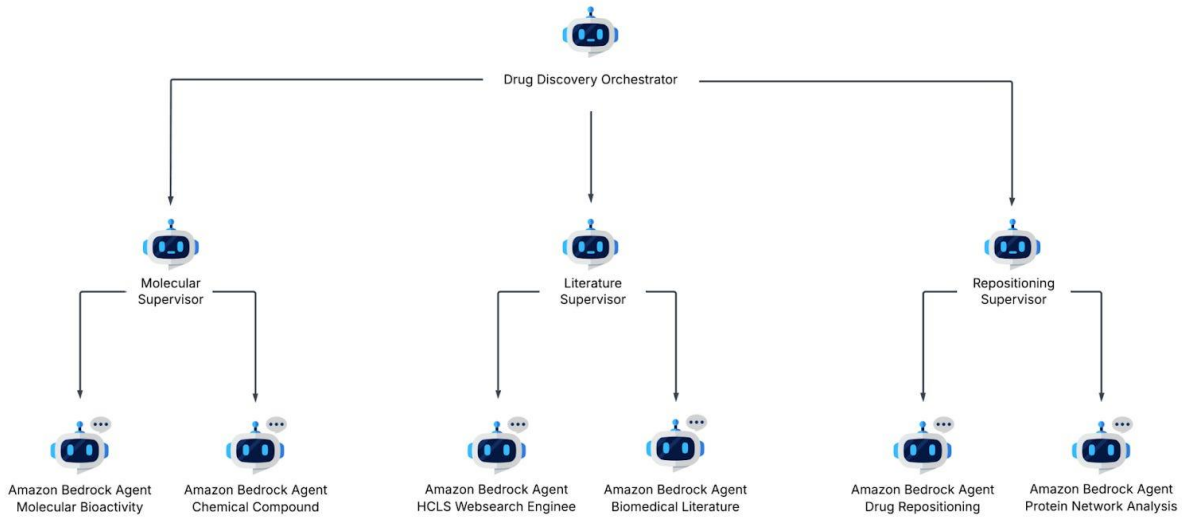
Visuals depicting agents competing for shared memory/resources.

A sequence diagram example for multi-agent orchestration and resource access.

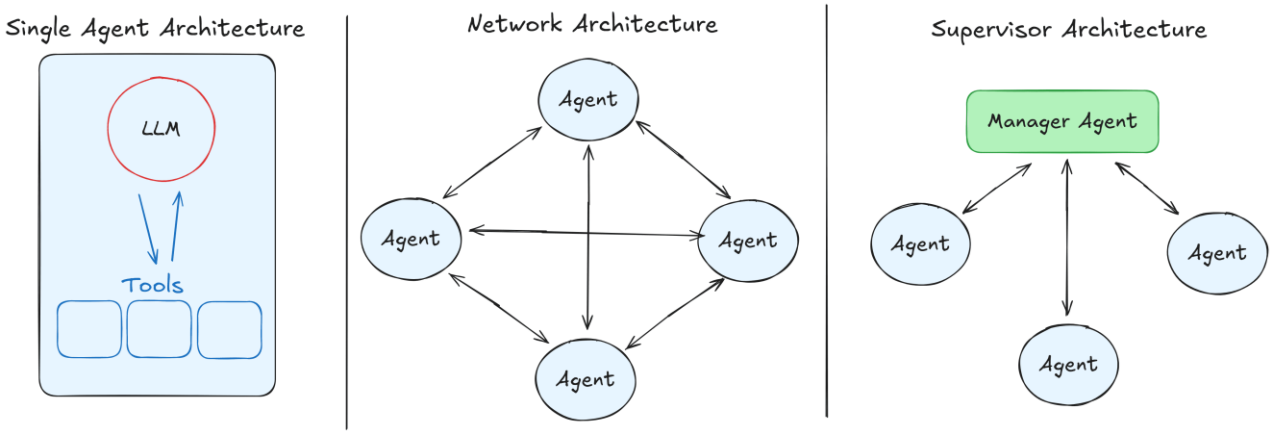
Connected agent graph showing interaction and connection protocols



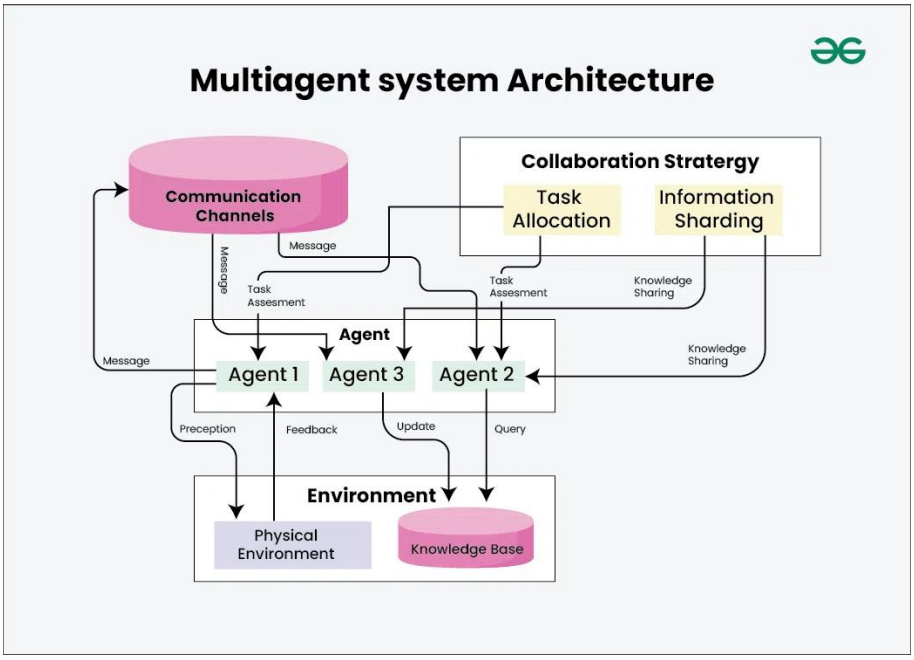
Deep Learning Pipeline
Diagrams of feature extraction, classification (like MobileNet), and scoring flows.
Principal Component Analysis pipeline for dimensionality reduction and feature handling, akin to your pre-processing. Image feature extraction workflow in computer vision, matching your sensor data to MobileNet step.



Arbitration Engine Internals
Close-ups on scheduling and priority mechanisms.
Shared memory multiprocessor architecture with rotating access, contrasting your dynamic confidence-based approach. AI agent core architecture steps including perception and decision-making



Control Flow Sequence
Time-based flows for arbitration cycles.
Detailed sequence diagram for agent requests, classification, and response merging in multi-agent setup. Multi-agent orchestration patterns like sequential and concurrent processing.



Multi-management system architecture uses a central orchestrator to route requests from multiple tenants or agents to shared resources like memory pools.A classifier or registry assigns tasks; arbitration (e.g., AI-driven like your Wild Bear) resolves conflicts based on priorities and confidence scores.
Data flows: inputs → processing → shared storage → merged outputs, with monitoring for scalability and isolation.
Benefits include cost-efficient resource pooling, dynamic allocation, and QoS in multi-agent or multi-tenant environments.

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